

Sigmoidal Function - 30 MCQ Questions

1. The range of the sigmoid function is:

- A) $(-\infty, +\infty)$
- B) $[0, 1]$
- C) $(-1, 1)$
- D) $(0, +\infty)$

Answer: B

2. If $\sigma(0)$ is computed for a logistic sigmoid, the result is:

- A) 0.0
- B) 0.25
- C) 0.5
- D) 1.0

Answer: C

3. The derivative of the sigmoid function $\sigma(x)$ can be expressed as:

- A) $\sigma'(x) = \sigma(x) \cdot (1 - \sigma(x))$
- B) $\sigma'(x) = \sigma(x)^2$
- C) $\sigma'(x) = 1 - \sigma(x)$
- D) $\sigma'(x) = e^{\sigma(x)}$

Answer: A

4. Which of the following is a limitation of the sigmoid function in deep neural networks?

- A) Gradient explosion
- B) Gradient vanishing
- C) Non-differentiability
- D) Output range too large

Answer: B

5. For $x \rightarrow +\infty$, $\sigma(x)$ approaches:

- A) 0
- B) 1
- C) ∞
- D) 0.5

Answer: B

6. The sigmoid function is most sensitive to changes in x when:

- A) $x \ll 0$
- B) $x \approx 0$
- C) $x \gg 0$
- D) At all values equally

Answer: B

7. In binary classification using logistic regression, the sigmoid output represents:

- A) The predicted class (0 or 1) directly
- B) The raw decision boundary
- C) The probability of the positive class
- D) The loss function

Answer: C

8. The output of $\sigma(-x)$ can be expressed in terms of $\sigma(x)$ as:

- A) $1 - \sigma(x)$
- B) $-\sigma(x)$
- C) $\sigma(x)$
- D) $e^{\sigma(x)}$

Answer: A

9. The point where the sigmoid has maximum slope is:

- A) $x = 0$
- B) $x = 1$
- C) $x = -1$
- D) $x \rightarrow \infty$

Answer: A

10. Sigmoid activation squashes inputs to the range:

- A) $(0, 1)$
- B) $(-1, 1)$
- C) $(0, \infty)$
- D) $(-\infty, \infty)$

Answer: A

11. $\sigma(2) = ?$

- A) 0.731
- B) 0.881
- C) 0.622
- D) 0.500

Answer: A

12. $\sigma(-2) = ?$

- A) 0.269
- B) 0.119
- C) 0.378
- D) 0.500

Answer: A

13. If $\sigma(x) = 0.9$, find x approximately.

- A) 1.0
- B) 1.5
- C) 2.197
- D) 0.693

Answer: C

14. $\sigma(4) = ?$

- A) 0.982
- B) 0.961
- C) 0.880
- D) 0.990

Answer: A

15. $\sigma(-4) = ?$

- A) 0.018
- B) 0.038
- C) 0.072
- D) 0.002

Answer: A

16. If $\sigma(x) = 0.5$, then x must be:

- A) -1
- B) 0
- C) 1
- D) 2

Answer: B

17. Derivative at $x = 0$:

- A) 0.25
- B) 0.50

- C) 0.75
- D) 1.00

Answer: A

18. For $x = 3$, the derivative $\sigma'(x) \approx ?$

- A) 0.047
- B) 0.105
- C) 0.213
- D) 0.500

Answer: A

19. If $\sigma(x) = 0.8$, $\sigma'(x) = ?$

- A) 0.16
- B) 0.20
- C) 0.08
- D) 0.24

Answer: A

20. The sum $\sigma(3) + \sigma(-3) = ?$

- A) 1.000
- B) 0.950
- C) 1.050
- D) 0.500

Answer: A

21. Sigmoid is preferred over step function because:

- A) Continuous and differentiable
- B) Easier to implement in hardware
- C) Output range is larger
- D) Gives only binary output

Answer: A

22. Which function has similar shape but different range $(-1, 1)$?

- A) ReLU
- B) tanh
- C) softmax
- D) linear

Answer: B

23. In neural networks, sigmoid is usually avoided in deep hidden layers because:

- A) It's slow to compute
- B) Causes vanishing gradients
- C) Has no derivative
- D) Outputs negative values

Answer: B

24. In logistic regression, the log-odds ratio is:

- A) $\ln(\sigma(x)/(1 - \sigma(x)))$
- B) $1/(1 + e^{\sigma(x)})$
- C) $\sigma'(x)$
- D) $e^{\sigma(x)}$

Answer: A

25. For extreme values of x , sigmoid behaves like:

- A) Linear function
- B) Step function
- C) ReLU
- D) Gaussian

Answer: B

26. The function $\sigma(ax)$ where $a > 1$ becomes:

- A) Steeper curve
- B) Flatter curve
- C) Same curve
- D) Shifted right

Answer: A

27. If $a < 1$ in $\sigma(ax)$, then:

- A) The curve is steeper
- B) The curve is flatter
- C) Output range changes
- D) Gradient increases

Answer: B

28. Sigmoid function output can be interpreted as:

- A) Frequency
- B) Probability
- C) Distance
- D) Loss

Answer: B

29. The logistic sigmoid function is symmetric about:

- A) $y = 0.5$
- B) $x = 0$
- C) Both A and B
- D) None

Answer: C

30. Sigmoid's slope at $x=0$ is maximum and equals:

- A) 0.25
- B) 0.5
- C) 1.0
- D) 2.0

Answer: A